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Syddansk Universitet

Home Assignment
DS839: Causal Analysis in Business Data Science
Wordcount:

[redacted]

The aim of this task is to figure out: "*What is the causal effect of marijuana consumption on an individual's health, compared to not using marijuana*?"

The health effects of marijuana consumption

According to World Health Organisation "Cannabis *is globally the most commonly used psychoactive substance under international control. In 2013, an estimated 181.8 million people aged 15−64 years used cannabis for nonmedical purposes globally (uncertainty estimates 128.5–232.1 million)"* (WHO, 2016). The research also states that there is a worrying increasing demand for treatment for cannabis use. Although there has been done fundamental research studies on cannabis use there might still be gaps in the general knowledge of how cannabis use can affect one's health. The reason lying in how difficult it can be to collect population data on an illegal drug.

This assignment would like to analyse the causal impact of marijuana consumption on health with the dataset from NHANES. For context The National Health and Nutrition Examination Survey, NHANES, is a comprehensive research program on the health and nutritional status of adults and children in the US. The data are collected through interview data and physical examinations to generate national health statistics. The research began in the early 1960's and has since transitioned into a continuous annual program. The survey examines a nationally representative sample of 5000 individuals across the country. According to NHANES the methods include interviews that cover demographics, socioeconomic status, diet and health history whereas the examinations include medical, dental and physiological measurements, along with laboratory tests (CDC, 2024).

Data exploration

The NHANES dataset has 10000 observations distributed between 76 variables. 5000 observations from 2009-2015 and 5000 observations from 2010-2011. The dataset provides information on age, gender, race, income, education, work, sex, alcohol, marijuana, smoking etc. Below is a snapshot of some of the given data, 11 variables, from the dataset.

```
      Age          AlcoholDay      PhysActive    Depressed    SleepTrouble      BMI                Education         Race3
 Min.   : 0.00   Min.   : 1.000   No  :3677   None   :5246   No  :5799    Min.   :12.88   8th Grade     : 451   Asian   : 288
 1st Qu.:17.00   1st Qu.: 1.000   Yes :4649   Several:1009   Yes :1973    1st Qu.:21.58   9 - 11th Grade: 888   Black   : 589
 Median :36.00   Median : 2.000   NA's:1674   Most   : 418   NA's:2228    Median :25.98   High School   :1517   Hispanic: 350
 Mean   :36.74   Mean   : 2.914               NA's   :3327                Mean   :26.66   Some College  :2267   Mexican : 480
 3rd Qu.:54.00   3rd Qu.: 3.000                                           3rd Qu.:30.89   College Grad  :2098   White   :3135
 Max.   :80.00   Max.   :82.000                                           Max.   :81.25   NA's          :2779   Other   : 158
                 NA's   :5086                                             NA's   :366                           NA's    :5000
 AgeFirstMarij   RegularMarij  AgeRegMarij
 Min.   : 1.00   No  :3575    Min.   : 5.00
 1st Qu.:15.00   Yes :1366    1st Qu.:15.00
 Median :16.00   NA's:5059    Median :17.00
 Mean   :17.02                Mean   :17.69
 3rd Qu.:19.00                3rd Qu.:19.00
 Max.   :48.00                Max.   :52.00
 NA's   :7109                 NA's   :8634
```

When looking into the variable Age. The column gives us details on the population. The youngest person involved is 0-year-old. 25% of respondents are 17 years old. The average age is 36.74 years old and the oldest is 80 years old.

The data also has a variable for participants who have tried marijuana reported for age group 18-59. Where 2049 participants haven't smoked marijuana and 2892 has. Resulting in 5059 NAs

The data are cross-sectional on health and nutrition of the American population and is collected in intervals, but each survey represents a snapshot of the population at a specific point in time, in this case 2009-2011. CDC mentions that "*NHANES data are not obtained using a simple random sample. Rather, a complex, multistage probability sampling design is used to select a sample representative of the civilian noninstitutionalized resident population of the United States. NHANES excludes all persons in supervised care or custody in institutional settings, all active-duty military personnel, active-duty family members living overseas, and any other U.S. citizens residing outside the 50 states and the District of Columbia*" (CDC, 2013). The treatment is therefore not randomly assigned. When looking at the dataset the variables doesn't suggest a randomized treatment. Variables like Gender, Depressed, Income or Physactive are self-reported, observed or measured and not assigned experimentally.

Conditional Independence Assumption (CIA) & DAG

Since our treatment assignment for marijuana intake isn't random with unobserved confounders associated. The predictive effect may differ from the true causal effect. In this project health differences may reflect on other factors such as unhealthy habits like drinking and poor diet, access to healthcare, genetic factors or socioeconomic issues and not marijuana intake itself. The dataset from NHANES does adjust for different covariates with medical variables like diabetes and depression but it doesn't state the populations access to healthcare. This can be relevant for the health outcome since a delayed or no treatment will indicate a worse observed health, but it would be based on false attribution to marijuana. Furthermore, it could be relevant to look at the genetic or physiological factors since one's genetic risks and mental health could increase the use of marijuana and one's overall health.

To study the impact of smoking marijuana on health. We are invested in the causal effect of marijuana use un an individual's health.

If we start out by assuming that marijuana has no causal effect on health. We can't observe that both outcomes can be true for the same person, so we need strong assumptions or randomization to estimate it.

$$Y(1) = health\ of\ individual\ who\ uses\ marijuana$$

$$Y(0) = health\ of\ individual\ who\ doesn't\ use\ marijuana$$

The mathematically formula is:

$$Y = Y(0) = Y(1)$$

$$\delta = E[Y(1)] - E[Y(0)] = 0$$

The causal effect would be zero. The formula measures the effect if all else were equal.

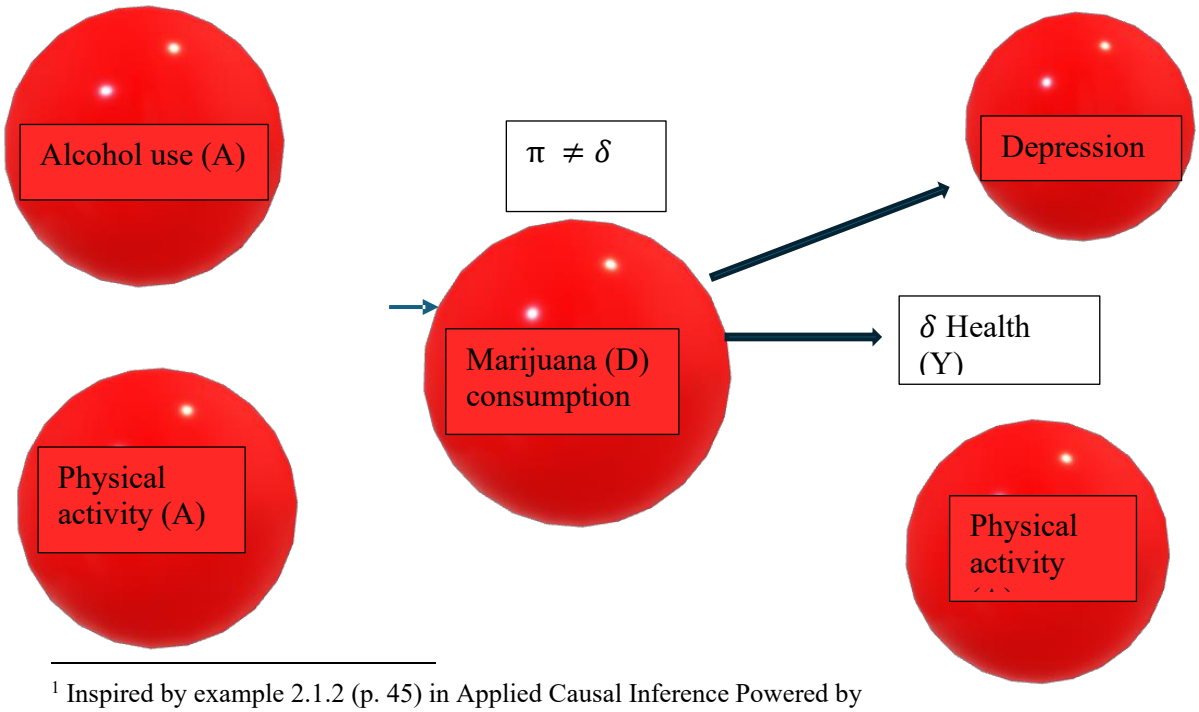
But in our observational data, marijuana intake (D = 1) isn't, as mentioned, randomly assigned. Therefore, individuals choose to consume it or not. In addition to that marijuana users may be more likely to engage in other unhealthy habits like alcohol intake, bad diet or less exercise. They might even use marijuana for its therapeutic effects on genetic or physiological factors. These confounders affect one's health, and they are correlated with use of marijuana. Therefore, the observed differences in health may reflect other factors π and not the true effect of cannabis itself δ

The formula will now look like

$$E[Y \mid D = 1] < E[Y \mid D = 0]$$

$$\pi = E[Y \mid D = 1] - E[Y \mid D = 0] < 0$$

It measures that the observered difference in health might be biased due to confounders[1]



[1] Inspired by example 2.1.2 (p. 45) in Applied Causal Inference Powered by ML and AI

The Directed Acyclic Graph illustrates the relation between marijuana consumption (D) and health (Y). It also highlights possible confounders like alcohol use (A) and other unhealthy traits like stress (U), that effect D and Y.

The causality we are interested in is whether marjiuana use affect health therefore:

$$\mathrm{D} \rightarrow \mathrm{Y}$$

U is for the unobserved confoundings since some may use marjiuana due to stress etc which also affect their health. Factors like stress are unmeasured and cant be controlled directly and violates the CIA:

$$U \rightarrow \mathrm{D} \text{ and } \mathrm{U} \rightarrow \mathrm{Y}$$

Alcohol is a measured confounder and there might be a correlation between use of alcohol and marijuana which also has a negative effect on health. This is why alcohol is included as a control variable:

$$A \rightarrow \mathrm{D} \text{ and } \mathrm{A} \rightarrow \mathrm{Y}$$

People with depression might use marijuana as self-medication and this effect health aswell. It's a confunder and must be controlled in the aim to avoid bias. Physical activity is also a confounder but it has a more positive effect on health and people who excercises may be less prone to use marijuana:

$$D <- \text{ Depression} \rightarrow \mathrm{Y} \text{ and } D <- \text{ Physical Activity} \rightarrow \mathrm{Y}$$

Depression, Alcohol and Physical activity are just some examples of confounders.

Table to compare mean

The purpose of this table is to summerize pre-treatment covariate balance treated and control units. The aim is to understand hiw much treated and control units differ in terms of observable characteristics

	Stratified by Marijuana			
	No	Yes	p	test SMD
n	2049	2892		
Age (mean (SD))	38.20 (11.98)	38.78 (12.12)	0.099	0.048
Gender = male (%)	972 (47.4)	1622 (56.1)	<0.001	0.174
Education (%)			<0.001	0.284
8th Grade	136 (7.1)	46 (1.7)		
9 - 11th Grade	218 (11.3)	337 (12.2)		
High School	381 (19.8)	591 (21.3)		
Some College	567 (29.4)	963 (34.7)		
College Grad	626 (32.5)	835 (30.1)		
HealthGen (%)			<0.001	0.187
Excellent	291 (14.2)	270 (9.4)		
Vgood	596 (29.1)	1031 (35.7)		
Good	823 (40.2)	1144 (39.6)		
Fair	293 (14.3)	385 (13.3)		
Poor	46 (2.2)	56 (1.9)		
AgeFirstMarij (mean (SD))	NaN (NA)	17.02 (3.90)	NA	NA
RegularMarij = Yes (%)	0 (0.0)	1366 (47.2)	<0.001	1.338
AgeRegMarij (mean (SD))	NaN (NA)	17.69 (4.81)	NA	NA
Depressed (%)			<0.001	0.173
None	1673 (81.6)	2159 (74.8)		
Several	253 (12.3)	524 (18.2)		
Most	123 (6.0)	203 (7.0)		
PhysActive = Yes (%)	1136 (55.4)	1715 (59.3)	0.007	0.078

The table informs that there out of the 10000 observations, we have 4941 non-missing valuse where 2049 individuals hasn't tried marijuana and 2892 has. Resulting in 5059 missing values. The table furthermore tells us that men are more likely to smoke marijuana with a standard mean difference on 0.174. A low SMD - close to zero, indicates better balance between the two groups.

Its noticably that marijuana users report more depressive symptoms according to the table with an SMD = 0.173 but it doesn't prove that marijuana is the cause of it. If individuals are more depressed on average then this factor alone lowers health score and might confound the effect of marijuana use on health unless its accounted for.

In addition to that the HealthGen shows an acceptable difference on health between users and non-users. A side note from the results that it may be relevant to adjust this task for gender, education and depression to isolate the causal effect of marijuanna use on health since these variables show significant imbalance between the two groups. When looking at Education we can see that users tend to have a different education levels. There is a higher rate of "some college" and "college grad" and lower of "8th grad". Education is a socioeconomic factor and it may influence habits and access to healthcare.

Linear Regression without control variables

To be able to run a linear regression of the outcome on the treatment, the variable HealthGen is treated as a numeric variable where 1 = Poor health and 5 = Excellent health.

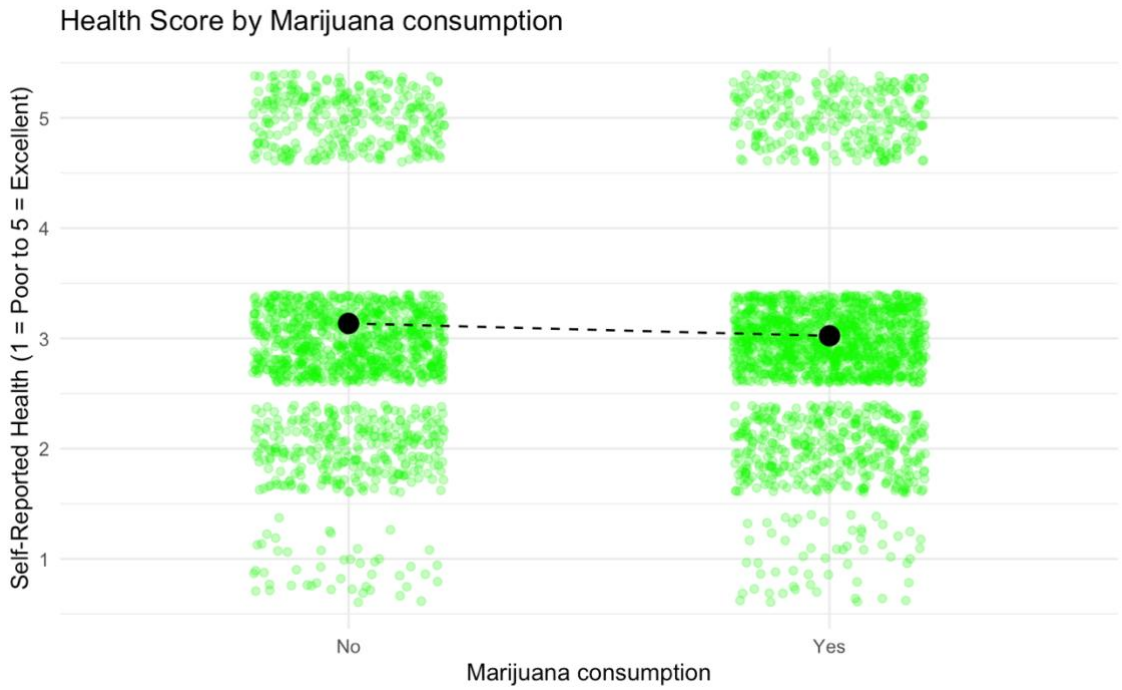
```
Call:
lm(formula = HealthGen_num ~ Marijuana, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.13558 -0.13558 -0.02318 -0.02318  1.97682 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   3.13558    0.02622  119.59  < 2e-16 ***
MarijuanaYes -0.11240    0.03501   -3.21  0.00134 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.9994 on 3306 degrees of freedom
  (6692 observations deleted due to missingness)
Multiple R-squared:  0.003108,	Adjusted R-squared:  0.002806 
F-statistic: 10.31 on 1 and 3306 DF,  p-value: 0.001339
```

The intercept is for those who answered “no” to Marijuana consumption. The non-users scored an estimated 3.14 on the average health score which is equivalent to “Good”. For marijuana users their score is on average 0.11 points lower as seen in the regression compared to non-users. The p-value is statistically significant with ($p = 0.001339$) and it tells us that there is strong evidence for the difference. When looking at marijuana and health without controlling for any other variable then marijuana consumption has a slightly lower self-reported health score, even though its statistically significant, the difference may be due to confounding hence we can’t interpret this as causal.



To vizualise the previous linear regres sion we make a plot for the group means with a regression line. The green dots in the plot represents the individual observations for the two groups. The two black points are the average health score for each group essentially Marijuana consumption no vs yes. We can see that the health score for non-users is slighty higher compared to users. The black dashed line shows the estimated linear effect between the group means.

Obs. Missing values are not included

Linear Regression with, without and interaction on control variables

	ols1 'no controls'	ols2 'with controls'	ols3 with controls interacted
Constant	3.136*** (0.0262)	4.041*** (0.1521)	4.798*** (0.2431)
MarijuanaYes	-0.1124** (0.0350)	-0.1013** (0.0384)	-0.3367 (0.3291)
MarijuanaYes x DepressedSeveral			0.0679 (0.1160)
MarijuanaYes x DepressedMost			-0.1401 (0.1983)
MarijuanaYes x Age			0.0082* (0.0034)
MarijuanaYes x AlcoholDay			0.0198 (0.0145)
MarijuanaYes x PhysActiveYes			0.0504 (0.0790)
MarijuanaYes x DaysMentHlthBad			0.0132* (0.0057)
MarijuanaYes x SleepTroubleYes			-0.0490 (0.0899)
MarijuanaYes x BMI			-0.0041 (0.0055)
MarijuanaYes x Education9-11thGrade			-0.3406 (0.2239)
MarijuanaYes x EducationHighSchool			-0.3365 (0.2232)
MarijuanaYes x EducationSomeCollege			0.1345 (0.2188)
MarijuanaYes x EducationCollegeGrad			0.1106 (0.2246)
MarijuanaYes x Race1Hispanic			0.0603 (0.1709)
MarijuanaYes x Race1Mexican			0.4905** (0.1556)
MarijuanaYes x Race1White			-0.0300 (0.1156)
MarijuanaYes x Race1Other			-0.0292 (0.1778)
MarijuanaYes x Gendermale			-0.2366** (0.0778)
MarijuanaYes x DaysPhysHlthBad			0.0098. (0.0052)
S.E. type	IID	IID	IID
Observations	3,308	2,405	2,405
R2	0.00311	0.25948	0.30890

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

In addition to our previous regression model we have added a controlled model and a interaction model. The OLS2 - Model with controls for Age, AlcoholDay, PhysActive etc. and even with these confounders marijuana consumption still has a negative influence on health. When looking at the model for interaction then the main effect of marijuana is non-significant since the effect of marijuana on average varies across subgroups. To point out some of the interactions we can look at race where mexican individuals health improves when influenced with marjiuana. The overall model shows that the use of marijuana is associated with lower self-reported health but the effect in magnitude suggest a limited relevance at population level. The results also indicate a robustness to confounders since the effects persists ($p < 0.05$) when adjusting for confounders. The interaction model shows that it could be relevant to prioritze subgroup effects over aggregated averages when studying the impact of treatment.

Alternative methods

Average Treatment Effect & Average Treatment Effect on the Treated

The alternative method from Lecture 2 is ATE & ATET and they will provide the causal estimates on the impact of marijuana consumption on health. The results are grouped by Age, Education, Depressed, Physactive and Gender.

ATE <dbl>	ATET <dbl>
-0.1271717	-0.0819064

1 row

The dataset includes 329 individuals and 10 variables. However, only 167 out of 329 individuals were suitable for incluison in the exact matching analysis when looking at marijuana user and non-users (n0 > 0 and n1 >0)

The Average Treatment Effect is estimated at -0.127 which suggests that if everyone in the given population consumed marijuana. It would have a negative influence on the average self-reported health score compared to if no one consumed it. The estimate is a result of matching different observable characteristics. In this case we used age, education, depression, physical activity and gender.

The Average Treatment Effect on the Treated also have a lower point score and is estimated at -0.082 which indicate that does who actually have used marijuana has a 0.082 lower point score compared to non-users even with the same observed treatments.

The given results tells us that marijuana has a negative influence on ones health when comparing between users and non-users and their self-reported health. Even after matching individuals with similar traits.

Augmented Inverse Probability Weighting

This method is used to estimate the average causal effects

```
Call:
lm(formula = beta_i ~ 1)

Residuals:
     Min       1Q   Median       3Q      Max 
-11.0935  -0.8499   0.0155   0.8521   6.7894 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept) -0.08215    0.03467   -2.37   0.0179 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.944 on 3142 degrees of freedom
```

The results from the AIPW Estimation shows an estimated ATE that is -0.082. The results has a statistical significant and it supports the idea of a relation between marijuana and impact on

health. This model suggests that users score 0.082 points lower when compared to non-users on health. It alligns with the results from the previous section for ATE and ATET.

Conditional Average Treatment Effect

I had issues with the package for CATE. My script couldn't run despite it being installed.

Discussion & conclusion

This section is a discussion and conclusion of the analysis of marijuana's influence on health with the use of NHANES data and different causal inference methods: ATE, ATET, AIPW, Three Model Linear regression, CIA, DAG & mean table. I would like to point out that the influence for marijuana consumption on health might would have looked different if I used the variable "RegularMarij" instead of "Marijuana". The participants from "RegularMarij" have been or are regular marijuana users and have consumped it at least once a month for year. Compared to the chosen variable "Marijuana" which is for those who have tried it. Consuming Marijuana regularly has a completely different impact on health, especailly if done over a longer period of time. The results from my analysis does show that marijuana consumers have a slightly lower score on their health score report compared to non-smokers, but the difference among the two group might have been more noticably.

When combining the results from multiple causal inference strategies it suggests that marijuana consumption is associated with a modest yet consistent reduction in self-reported health scores**.** The ATE indicates a decline estimated at -0.127 in health if everyone in the population consumed Marijuana. The effects remain even when adjusting for confounding factors such as Age, Education, Gender, Depressed and Activity. The validity of results across methods enhances credibility. Unobserved confoundings can never be ruled out entirely and the results should be interpreted with cautious causality. Among those who actually use marijuana the estimated reduction in health was 0.082 so they are actually less affected on average. But the results from ATE & ATET shows that marijuana does have a negative effect on health even after exact matching by age, edcuction etc. In addition to that the AIPW estimate -0.082 and alligns with the ATET. It could have been good to use CATE and capture the patterns also to provide a more personalized insights into the treatment effects and adjust to subgroups.

The answer to: "*What is the causal effect of marijuana consumption on an individual's health, compared to not using marijuana*?" is that the consumption of marijuana has a slighly negative

yet consisent causal effect on self-reported health. This can be concluded out from different methods use and how the findings hold when adjusting for confounders.

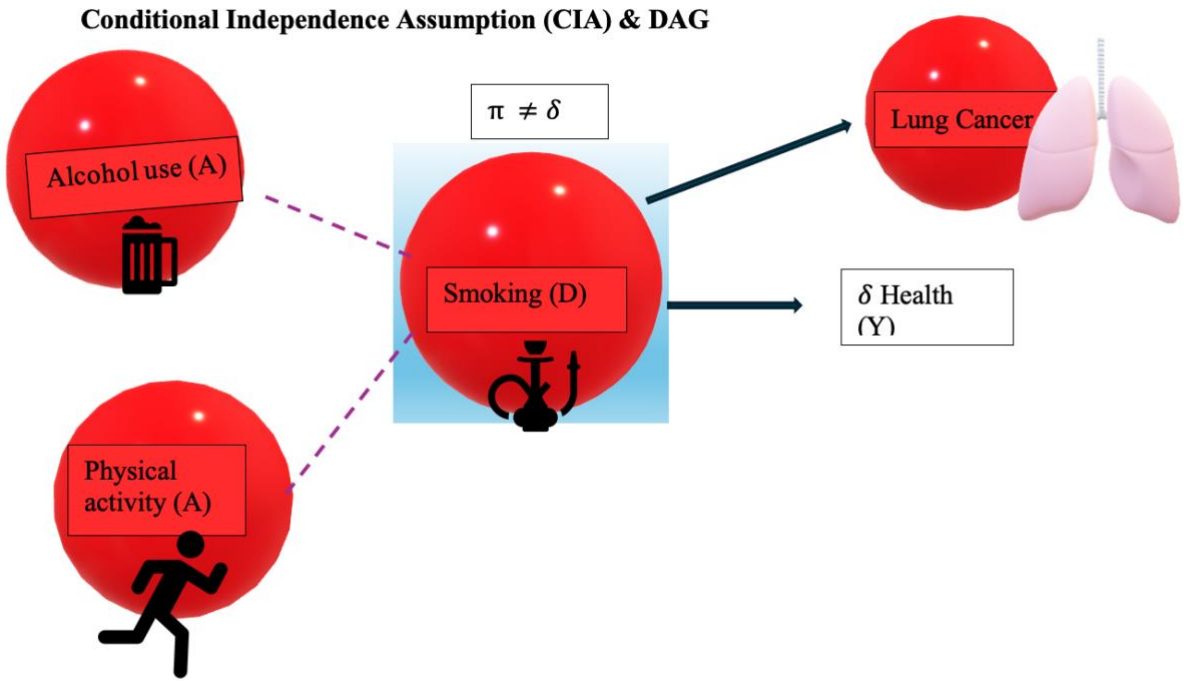
The aim of this task is to figure out: "*What is the causal effect of having smoked cigarettes on an individual's health, compared to not smoking*?"

The health effects of smoking cigarettes

Data

```
 SmokeNow     Smoke100       SmokeAge
 No   :1745   No   :4024   Min.   : 6.00
 Yes  :1466   Yes  :3211   1st Qu.:15.00
 NA's:6789    NA's:2765    Median :17.00
                           Mean   :17.83
                           3rd Qu.:19.00
                           Max.   :72.00
                           NA's   :6920
>
```

Within the dataset there is three variables regarding smoking. We have the variable "SmokeNow", which include study participants who are currently smoking cigarettes reguarly. Where 1466 participants answered "yes" to smoking now, 1745 answered no and the remaining 6789 subjects who haven't smoked 100 or more cigerettes are listed as NA. There is also the variable for "Smoke100" for those who have smoked at least 100 cigarettes. The last variable is for the age where the participants started to smoke on a regular basis. Where the youngest age is 6, average age is 17.83 and the oldest age is 72.



The CIA and DAG is similar to the previous one on Marijuana and Health.

Linear Regression without control variables

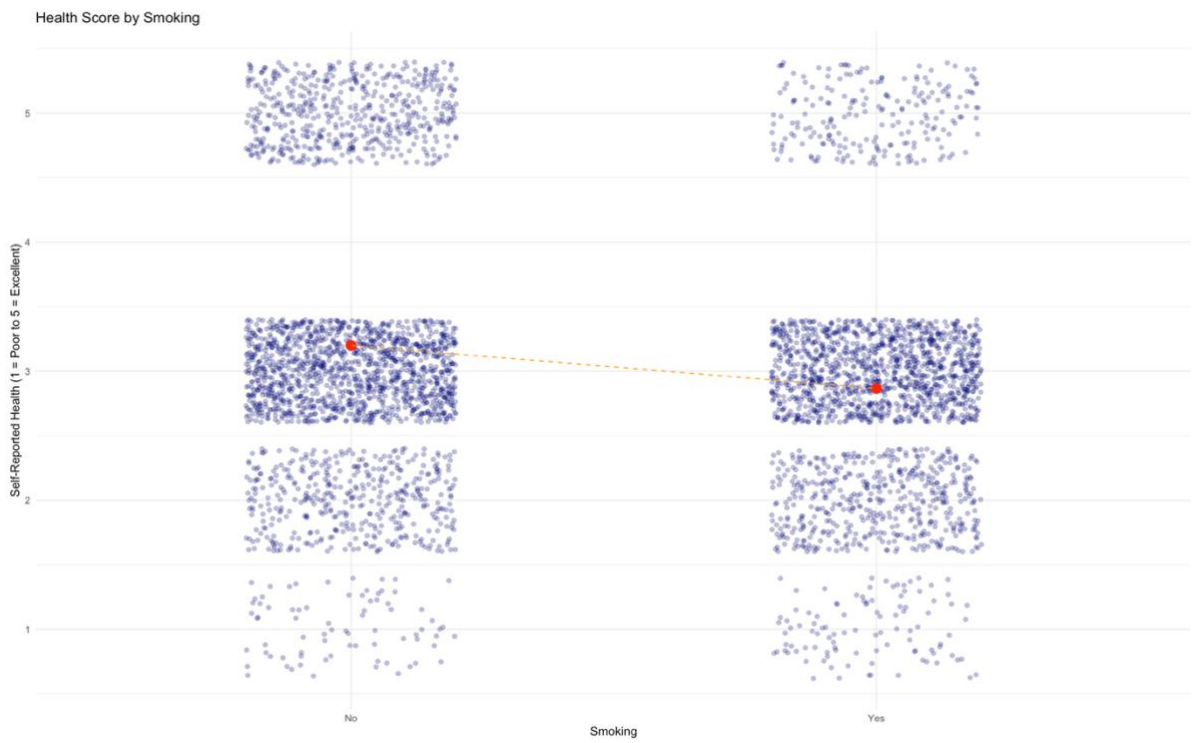
```
Call:
lm(formula = HealthGen_num ~ Smoke100, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.1995 -0.8657 -0.1995  0.1343  2.1343 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.19949    0.02082  153.69   <2e-16 ***
Smoke100Yes -0.33383    0.03050  -10.95   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.008 on 4391 degrees of freedom
  (5607 observations deleted due to missingness)
Multiple R-squared:  0.02656,	Adjusted R-squared:  0.02634 
F-statistic: 119.8 on 1 and 4391 DF,  p-value: < 2.2e-16
```

The intercept is based on none-smokers and there score is estimated to 3.19 on the health score. Slightly higher than the score smokers who on average scored 0.33 points lower. Based on this regression without control variables for smokers vs non-smokers and marijuana users vs none. The non-smokers has the best health.



To vizualise the previous linear regres sion we make a plot for the group means with a regression line. The navyblue dots in the plot represents the individual observations for the two groups. The two red points are the average health score for each group essentially smokers vs non. The is a bigger difference here than between marijuana-smokers vs non. Since the non-smokers have a higher health score in general and smokers health score is relatively lower when compared with non-smokers & marijuana-users.

Linear Regression with, without and interaction on control variables

	ols1 'no controls'	ols2 'with controls'	ols3 with controls interacted
Constant	3.199*** (0.0208)	3.793*** (0.1285)	4.524*** (0.1970)
Smoke100Yes	-0.3338*** (0.0305)	-0.2266*** (0.0333)	-1.035*** (0.2586)
Smoke100Yes x DepressedSeveral			0.3170** (0.0970)
Smoke100Yes x DepressedMost			0.2392 (0.1614)
Smoke100Yes x Age			-7.5e-5 (0.0021)
Smoke100Yes x AlcoholDay			0.0450*** (0.0125)
Smoke100Yes x PhysActiveYes			0.1248. (0.0676)
Smoke100Yes x Education9-11thGrade			-0.0915 (0.1712)
Smoke100Yes x EducationHighSchool			-0.1931 (0.1632)
Smoke100Yes x EducationSomeCollege			-0.1388 (0.1613)
Smoke100Yes x EducationCollegeGrad			0.2103 (0.1667)
Smoke100Yes x Gendermale			-0.1525* (0.0657)
Smoke100Yes x DaysMentHlthBad			-0.0012 (0.0047)
Smoke100Yes x SleepTroubleYes			0.0717 (0.0731)
Smoke100Yes x BMI			0.0244*** (0.0047)
Smoke100Yes x Race1Hispanic			0.0576 (0.1560)
Smoke100Yes x Race1Mexican			0.1652 (0.1411)
Smoke100Yes x Race1White			-0.1379 (0.0978)
Smoke100Yes x Race1Other			0.2541 (0.1583)
Smoke100Yes x DaysPhysHlthBad			0.0051 (0.0041)
S.E. type	IID	IID	IID
Observations	4,393	3,118	3,106
R2	0.02656	0.27716	0.32338

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Through out the three models there is a consistent indicator for smoking being causally linked to worse self-reported health. Furthermore we can see that smoking affects differently and different people based on covariates like Gender

Average Treatment Effect & Average Treatment Effect on the Treated

ATE <dbl>	ATET <dbl>
-0.1776273	-0.1755104

1 row

The dataset includes 329 individuals and 10 variables. However, only 167 out of 329 individuals were suitable for incluison in the exact matching analysis when looking at marijuana user and non-users ($n0 > 0$ and $n1 > 0$)

The Average Treatment Effect is estimated at -0.177 which indicates that if everyone in the given population smoked cigarettes then it would have a negative influence on the average self-reported health score and more negative than for marijuana user. The estimate is including the result of matching different characteristics.
The ATET on the Treated is estimated at -0.175 which is close to the ATE.
The outcome suggets that smoking has a negative influence on ones health. And it the influence is greater than the one for marijuana users vs non-users.

Discussion & Conclusion

Through out the analysis for smoked cigarettes and health. There has been estimated and proved the causal effect of smoking and the impact has had an negative relation to ones self-reported health even when adjusting to observed confounders. The different outcomes from the methods has shown a consistent and negative effect on health. The answer to "*What is the causal effect of having smoked cigarettes on an individual's health, compared to not smoking*?" is smoking causally reduces one's health.

References

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Chernozhukov Victor & co. (2025). Applied Causal Inference Powered by ML and AI. https://chapters.causalml-book.org/CausalML_book_2022.pdf